The replicators

***p. 18** The simplified account I shall give [of the origin of life] is probably not too far from the truth.*

There are many theories of the origin of life. Rather than labour through them, in *The Selfish Gene* I chose just one to illustrate the main idea. But I wouldn't wish to give the impression that this was the only serious candidate, or even the best one. Indeed, in *The Blind Watchmaker*, I deliberately chose a different one for the same purpose, A. G. Cairns-Smith's clay theory. In neither book did I commit myself to the particular hypothesis chosen. If I wrote another book I should probably take the opportunity to try to explain yet another viewpoint, that of the German mathematical chemist Manfred Eigen and his colleagues. What I am always trying to get over is something about the fundamental properties that must lie at the heart of any good theory of the origin of life on any planet, notably the idea of self-replicating genetic entities.

***p. 21** 'Behold a virgin shall conceive … '*

Several distressed correspondents have queried the mistranslation of 'young woman' into 'virgin' in the biblical prophecy, and have demanded a reply from me. Hurting religious sensibilities is a perilous business these days, so I had better oblige. Actually it is a pleasure, for scientists can't often get satisfyingly dusty in the library indulging in a real academic footnote. The point is in fact well known to biblical scholars, and not disputed by them. The Hebrew word in Isaiah is עלמה(*almah*), which undisputedly means 'young woman', with no implication of virginity. If 'virgin' had been intended, בתולה(*bethulah*) could have been used instead (the ambiguous English word 'maiden' illustrates how easy it can be to slide between the two meanings). The 'mutation' occurred when the pre-Christian Greek translation known as the Septuagint rendered *almah* into παθθένος (parthenos), which really does usually mean virgin. Matthew (not, of course, the Apostle and contemporary of Jesus, but the gospel-maker writing long afterwards) quoted Isaiah in what seems to be a derivative of the Septuagint version (all but two of the fifteen Greek words are identical) when he said, 'Now all this was done, that it might be fulfilled which was spoken of the Lord by the prophet, saying, Behold, a virgin shall be with child, and shall bring forth a son, and they shall call his name Emmanuel' (Authorized English translation). It is widely accepted among Christian scholars that the story of the virgin birth of Jesus was a late interpolation, put in presumably by Greek-speaking disciples in order that the (mistranslated) prophecy should be seen to be fulfilled. Modern versions such as the *New English Bible* correctly give 'young woman' in Isaiah. They equally correctly leave 'virgin' in Matthew, since there they are translating from his Greek.

***p. 25** Now they swarm in huge colonies, safe inside gigantic lumbering robots …*

This purple passage (a rare—well, fairly rare—indulgence) has been quoted and requoted in gleeful evidence of my rabid 'genetic determinism'. Part of the problem lies with the popular, but erroneous, associations of the word 'robot'. We are in the golden age of electronics, and robots are no longer rigidly inflexible morons but are capable of learning, intelligence, and creativity. Ironically, even as long ago as 1920 when Karel Capek coined the word, 'robots' were mechanical beings that ended up with human feelings, like falling in love. People who think that robots are by definition more 'deterministic' than human beings are muddled (unless they are religious, in which case they might consistently hold that humans have some divine gift of free will denied to mere machines). If, like most of the critics of my 'lumbering robot' passage, you are not religious, then face up to the following question: What on earth do you think you are, if